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The effects of knee ligament load using simulated hip abductor and hamstring muscle strengthening during cutting maneuver

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Abstract

Background: This study aimed to analyze knee ligament of load and joint moment to simulate the strengthening of the hip abductor and hamstring muscles using musculoskeletal modeling, thereby contributing to decrease of knee ligament load.

Methods: Forty participants (age: 21.85 ± 1.90 years; height: 1.76 ± 0.06 m; body mass: 68.5 ± 7.06 kg) were instructed to run at 4.5 ± 0.2 m/s from a 5 m distance posterior to the force plate, land their feet on the force plate, and perform the cutting maneuver on the left. In the musculoskeletal modeling, the hip abductor and hamstring muscles were targeted to construct a model with a 30% increase in the contraction force of the hip abductor, hamstring, and both 2 muscles. The variables were the ligament force and knee joint moment. One-way repeated measure ANOVA and Bonferroni test were used to compare the *abductor/hamstring, abductor, hamstring and control* models.

Results: There were significant differences in anterior bundle of the anterior cruciate ligament (ACL) (P = .001), inferior bundle of the superficial layer of the medial collateral ligament (MCL) (P = .016), and posterior bundle of the superficial layer of the MCL (P = .022) force. The post hoc showed that the *hamstring* model had lower anterior bundle of the ACL and inferior bundle of the superficial layer of the MCL than the *abductor/hamstring* and *abductor* models (P < .05) and lower posterior bundle of the superficial layer of the MCL than the *abductor* and *control* models (P < .05). There was a significant difference in the adduction (P = .028) and internal rotation moments (P = .014). The post hoc showed that both moments were lower in the *hamstring* model than in the other models (P < .05).

Conclusions: The hamstring strengthening may contribute significantly to preventing ACL or MCL injury by reducing knee ligament load.

Abbreviations: aACL = anterior bundle of the ACL, ACL = anterior cruciate ligament, CMC = computed muscle control, FD = forward dynamics, GRF = ground reaction force, IK = inverse kinematics, iMCL = inferior bundle of the superficial layer of the MCL, MCL = medial collateral ligament, pMCL = posterior bundle of the superficial layer of the MCL.

Keywords: cutting maneuver, injury prevention, knee joint, knee ligament, musculoskeletal modeling

1. Introduction

The cutting motion reduces the speed of running to change direction, which causes a high load on the knee joint.[1,2] The major areas of injury within the knee joint are the anterior cruciate ligament (ACL) and the medial collateral ligament (MCL). The ACL injury can be divided into contact injury due to an external shock and non-contact injury without a physical contact, and approximately 70% of ACL injury is accounted for by non-contact injury.[3] The MCL increases the internal stability of the knee joint by reducing the knee joint abduction, femur protraction and internal rotation upon the shock of the foot contacting the ground.[4,5] The MCL is relatively low in durability so that approximately 1/3 of ACL injury is accompanied with MCL and medial meniscus injuries.[6,7]

During cutting maneuver, the knee joint muscles and ligaments are known to undergo transient strains.[6] In the perspective of knee joint kinetics, the strain and load on the ACL have been reported to increase as a result of various factors

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including knee flexion and abduction movements, shear force, adduction and internal rotation moments, the quadriceps and the hamstring muscles fatigue and peak vertical ground reaction force (GRF).[8,9] Among the lower limb muscles, the hip abductor and hamstring muscles play crucial roles during cutting maneuver so that both these muscles should be considered in analyzing the knee joint biomechanics.[10] When the muscles of the femoral quadriceps contract at an angle less than 60 degrees of flexion of the knee joint, the ACL is damaged by moving forward from the tibia, but the strong contraction of the hamstring muscles increases the force applied backward to protect the ACL.[11,12] Several previous studies reported that the decrease in the contraction force of hip abductor and hamstring muscles could increase the knee joint flexion and external rotation moments as well as the extension and internal rotation moments.

The conventional training methods performed by athletes do not readily allow the hip abductor and hamstring muscles to be independently strengthened. To overcome the limitation, it is necessary to clearly determine the effects on the kinetic parameters of the knee joint by developing a novel musculoskeletal model where the contraction force of each muscle can be increased. This study aimed to simulate the strengthening of hip abductor and hamstring contraction forces through musculoskeletal modeling and compare the knee joint load and kinetic parameters in the cutting maneuver, with an aim to contribute to decrease of knee ligament load. For this study, the following hypotheses were proposed: During the cutting maneuver, strengthening the hamstring muscle group is expected to reduce the load on the knee ligaments, and a reduction in the adduction and internal rotation and internal rotation moments of the knee joint is predicted.

2. Methods

2.1. Participants

Participants in this study were 40 healthy males (age: 21.85 ± 1.90 years, height: 1.76 ± 0.06 m, body mass: 68.5 ± 7.06 kg). Each participant was right-foot dominant, participating in diverse sports activities three times a week for an average duration of 1 to 2 hours daily.[8] For inclusion in the study, participants needed to have no history of lower extremity injuries and no knee injuries in the preceding 6 months. Participants who did not meet these criteria were not included in the study.[13] The data of all participants were anonymized for privacy protection, and the relevant details were approved by the Institutional Review Board prior to conducting the data analysis (KNUE-202108-ETC-0305-01).

2.2. Equipments

Sixteen infrared cameras (Oqus 7+; Qualisys, SWE) were used for 3D motion analysis and one force plate (Type 9260AA6; Kistler, SWI) was used to obtain the kinematic and kinetic data. Following previous studies, reflective markers were attached to the major joints and nodes of the upper and lower limbs of the participants.[14,15] The 40 reflective markers had joint markers attached to the anterior superior iliac spine, posterior superior iliac spine, sacrum, shoulder, elbow, hip, knee and ankle joint and tracking markers attached to the head, trunk, upper arm, lower arm, thigh, shank and foot.[14,15] To define the position and orientation of the camera, the infrared cameras were calibrated, and the sampling rate was set as 150 frames/sec. The sampling rate of the force plate was set as 1500 Hz.

2.3. Experimental procedure

All participants wore the same experimental clothes and shoes, with reflective markers (1.2 cm diameter) attached to the major joints of the body. For the cutting maneuver, the participants were instructed to start at 5 m posterior to the force plate and run at 4.5 ± 0.2 m/s (Fig. 1). While the right foot landed on the force plate in the forward direction, only the accurate motion in the 35°–55° range on the left of the direction of motion was examined for triplicate measurements.[13,16] To control the set speed, the timing gait (Witty, Microgate, Italy) was installed. The first gate was installed at a 3.5 m point on the 5 m linear course for the cutting maneuver and the second gate was installed 1.5 m in the back of the force plate, while the speed between the first and second gates was measured (Fig. 1).

2.4. Data analysis

Minimum sample size was estimated using G*Power3 software for an alpha level of 0.05 and a beta of 0.80. Sample size calculations were based on differences between healthy adults and those with cutting maneuver reported in the literature. In the Visual 3D software, the cutoff frequency was set as 15 Hz, and

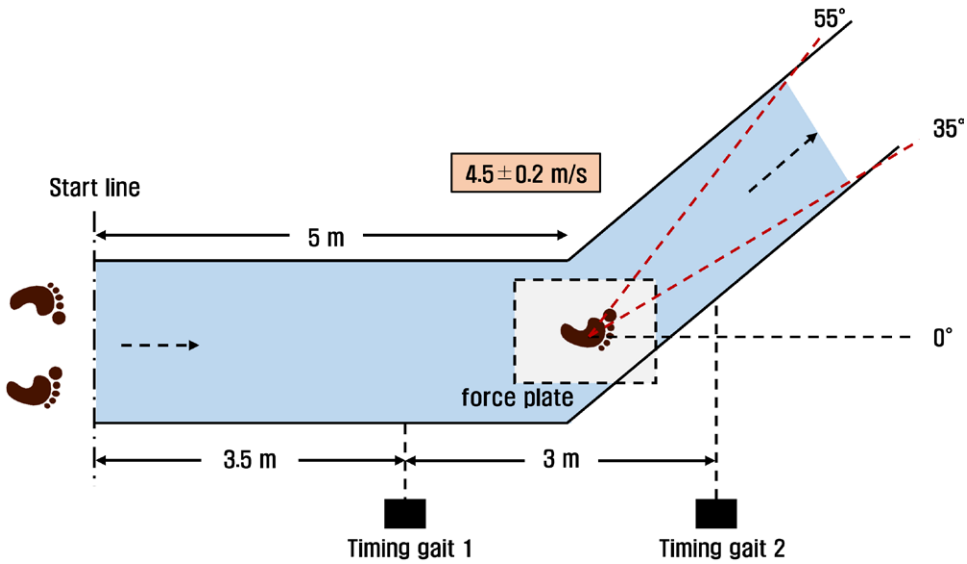

Figure 1. Diagram of the cutting task.

the 4th order Butterworth low-pass filter was applied to the raw data of the 3-dimensional position and GRF.[17] In this study, the analysis was focused on the phase of the passive force on the body upon landing (passive landing phase), the definition of which was from the feet landing on the ground to the maximum knee flexion.[18]

2.4.1. Knee ligament and muscle modeling. For the knee ligament and muscle modeling, OpenSim (ver. 3.3; Stanford University, Stanford, CA) was used. OpenSim is a program to estimate the variables based on the input of the muscle, ligament and kinetics data.[5,19] The musculoskeletal model used in the OpenSim program was the *gait2392* developed by Delp et al.[20] The model comprised 14 segments, 23 degrees of freedom and 92 muscle actuators, among which the hip abductor group consisted of sartorius, gluteus medius, gluteus minimus and tensor fasciae latae[21] and the hamstring group consisted of biceps femoris, semitendinosus and semimembranosus.[22] The model with increased muscle contraction forces was applied in analyzing the load on the knee ligament in this study; the model was developed for the ACL (2-bundles) and the MCL (5-bundles) with a high frequency of injury during cutting maneuver.[23] The max isometric force was increased by 30% for hip abductor and hamstring groups in the following models: an *abductor/hamstring* model with the hip abductor and hamstring contraction forces concurrently increased by 30%; an *abductor* model with the hip abductor contraction force increased by 30%; and a *hamstring* model with the hamstring contraction force increased by 30% (Fig. 2).[24] And the *control* model was defined as not strengthening the 2 muscle groups.

2.4.2. OpenSIM data process. The process of extracting the knee joint data in this study consisted of body scaling, inverse kinematics (IK), inverse dynamics, residual reduction algorithm, computed muscle control (CMC), and forward dynamics (FD).[8,25] First, the data of the physical composition of the participants were applied in body scaling. For IK, the kinematics data were obtained based on the coordinates of all markers. For inverse dynamics, the force and moment of each joint were estimated based on the IK data. For residual reduction algorithm, the force and moment of each joint were estimated by minimizing the measurement errors; to minimize the errors across the IK and GRF data, the mass was rearranged across the nodes, and the errors of the estimated Max Residual Force and Max Residual Moment fell within the GOOD and OKAY range to indicate that the data reliability was high (Table 1).

For CMC, the individual muscle excitation is estimated in consideration of the maximum muscle force and the actual measured values of motion and force. The CMC data is applied in calculating the individual ligament length and active force through the FD.[25]

The Hill model was used to calculate the ACL and MCL forces, by the following equation.[26]

$$f_m^* = [a_m^* f_{lv}(l_m^*, i_m^*) + f_{psv}(l_m^*)]\cos(a_m^*)$$

f_m^* = ligament force, a_m^* = muscle activation, f_{lv} = the dynamic force effected by the force-length-velocity curve of the Hill model, l_m^* = the muscle length, i_m^* = tendon velocity acting in the muscle direction, f_{psv} = passive force, and a_m^* = the muscle pennation angle.

The calculated values of ACL and MCL forces in time-series data were normalized using the participants' weight data, after which the maximum value (max ligament force) in passive landing phase was estimated. The knee joint moment in the sagittal, frontal and transverse planes was extracted as the time-series data based on the motion reproduced in the FD. The maximum value in passive landing phase was estimated after the normalization using the participants' height and weight data.[27]

2.5. Statistical analyses

Prior to statistical analyses, all estimated variables were tested for normality via one-sample Kolmogorov–Smirnov test. The result confirmed the normality of all variables so that the kinetics data were statistically analyzed via one-way repeated measure ANOVA. The post hoc test on the variation across the models was based on the Bonferroni correlation. Effect sizes were quantified as Cohen's *d*. All statistical analyses were performed using the SPSS 23.0 (SPSS Inc., Chicago, IL) with the level of significance set as $\alpha = .05$.

3. Results

3.1. Ligament force

The result of analyzing the ligament force in response to the increased muscle force during cutting maneuver is presented in Table 2. There were significant differences in anterior bundle of the ACL (aACL) ($F = 5.858$, $P = .001$), inferior bundle of the superficial layer of the MCL (iMCL) ($F = 3.991$, $P = .016$), and posterior bundle of the superficial layer of the MCL (pMCL) ($F = 3.475$, $P = .022$). The post hoc test showed that the *hamstring* model had lower aACL and iMCL than the *abductor/hamstring* and *abductor* models ($P < .05$) and lower pMCL than the *abductor* and *control* models ($P < .05$).

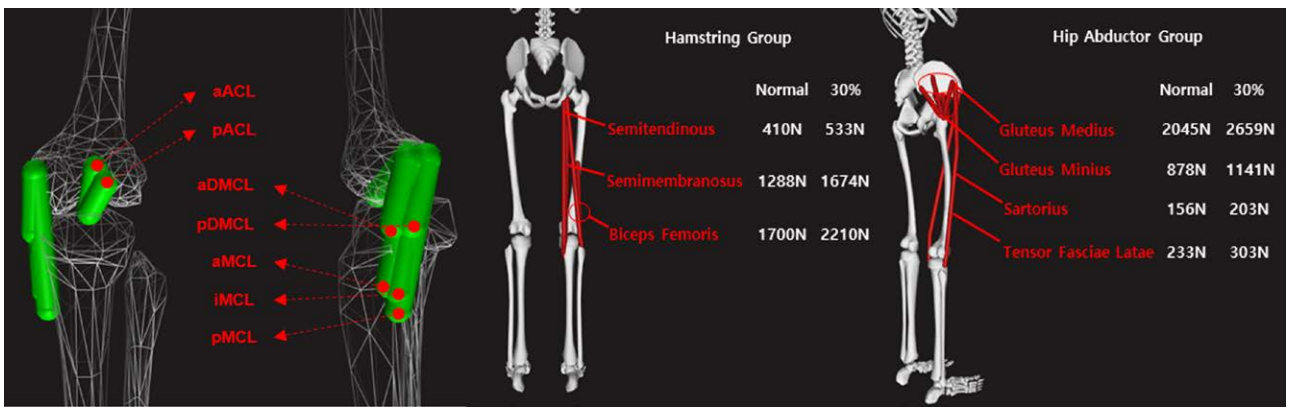

Figure 2. Figure 1. (1) The attachment sites of knee ligament bundles (left picture). (2) No contraction (normal) and 30% contraction of hamstring muscle group (center picture), and (3) no contraction and 30% contraction of hip abductor muscle group (right picture). aACL = anterior bundle of the ACL, aDMCL = anterior bundle of the deep layer of the MCL, aMCL = anterior bundle of the superficial layer of the MCL, iMCL = inferior bundle of the superficial layer of the MCL, pACL = posterior bundle of the ACL, pDMCL = posterior bundle of the deep layer of the MCL, pMCL = posterior bundle of the superficial layer of the MCL.[23]

Table 1

Mean of root-mean-squared residual forces (N) and moments (Nm) produced during RRA in OpenSim.

	Abductor/hamstring	Abductor	Hamstring	Control
FX	4.95 ± 7.29	8.69 ± 8.79	5.53 ± 9.02	5.22 ± 7.96
FY	16.77 ± 4.99	16.64 ± 7.31	20.98 ± 4.82	19.20 ± 2.05
FZ	4.69 ± 6.06	4.21 ± 4.52	5.91 ± 6.71	12.68 ± 2.57
MX	7.44 ± 8.56	13.61 ± 3.57	14.82 ± 8.90	25.13 ± 5.54
MY	10.74 ± 3.70	7.45 ± 2.74	13.81 ± 4.21	1.38 ± 0.87
MZ	3.89 ± 3.40	10.43 ± 1.27	5.28 ± 2.12	5.41 ± 2.34

Table 2

Results of knee ligaments force as strengthening difference during cutting.

Unit: N/BW

	Abductor/hamstring	Abductor	Hamstring	Control	*F*	*P*	Cohen's *d*
aACL	1.890 ± 0.339†	1.866 ± 0.436‡	1.656 ± 0.512%	1.815 ± 0.407	5.858	**.001***	1.31
pACL	1.234 ± 0.476	1.215 ± 0.393	1.302 ± 0.472	1.170 ± 0.414	1.625	.200	0.40
aMCL	0.732 ± 0.961	0.684 ± 0.890	0.674 ± 0.885	0.826 ± 1.255	0.654	.485	0.16
iMCL	1.007 ± 0.739	1.054 ± 0.686§	0.817 ± 0.568‖	1.088 ± 0.726	3.991	**.016***	0.93
pMCL	0.724 ± 0.619†	0.750 ± 0.609‡	0.536 ± 0.489	0.594 ± 0.644	3.475	**.022***	0.82
aDMCL	0.631 ± 0.781	0.710 ± 0.774	0.603 ± 0.761	0.648 ± 0.814	1.500	.230	0.37
pDMCL	1.210 ± 0.828	1.161 ± 0.762	1.078 ± 0.751	1.291 ± 1.130	1.795	.173	0.44

*$P < .05$.

†Significant difference between abductor/hamstring and hamstring.

‡Significant difference between abductor/hamstring and control.

§Significant difference between abductor and hamstring.

‖Significant difference between hamstring and control.

3.2. Knee moment

The result of analyzing the joint moment in response to the increased knee moment during cutting maneuver is shown in Table 3. There was a significant difference in the adduction ($F = 3.775$, $P = .028$) and internal rotation ($F = 3.698$, $P = .014$) moments. The post hoc test showed that both moments were lower in the *hamstring* model than in the other models ($P < .05$).

4. Discussion

In this study, the effects of increasing the contraction force of hip abductor and hamstring muscles on the kinetic parameters of the knee joint in the cutting maneuver were determined through computer simulation. For this, an *abductor/hamstring* model with 30% increased hip abductor and hamstring contraction forces, an *abductor* model with 30% increased hip abductor contraction force and a *hamstring* model with 30% increased hamstring contraction force were developed. The data of cutting maneuver performed by the participants were used in analyzing the maximum ligament force, and maximum knee joint moment.

The forces of aACL, iMCL, and pMCL during cutting maneuver were significantly lower in the *hamstring* model than in the other models ($P < .05$). In a previous study, the lower limb muscle activity during cutting maneuver was compared between a group with a history of knee ligament injury and the group without any past injury.[28] It was shown that although there was no difference in the quadriceps between injury and no-injury groups, the hamstring muscle activity in the injury group was an approximately 5% to 7% lower than no-injury group. The can be due to the difference in muscle strength between the groups, resulting from the history of injury, whereby the low quadriceps/hamstring ratio could limit the cutting maneuver.[29] In this study, the difference in aACL force in the hamstring model was lower about 10% than the abductor model. In addition, the abductor/hamstring model is estimated to have a low Q/H ratio because the 2 muscles were equally strengthened, which did not significantly impact reducing the load on the knee ligament. This means that the function of the hamstring muscle is relatively reduced due to the contraction of the abductor muscle group, and the muscle activation of the hamstring group was decreased compared to the abductor muscle during the cutting maneuver. Strengthening of the Hamstring muscles reduces the anterior shear force of the knee joint by about 15% during cutting, and the hamstring muscles are known to be able to distribute the load to the hip and ankle joints when supporting weight.[1,24,30]

The results in this study showed that, among the 5 bundles of MCL during cutting maneuver, the posterior bundles had the load reduced. We found that the iMCL of the hamstring model decreased by 25%, and the pMCL decreased by 33% compared to the abductor model during cutting. This means that hamstring muscles have the function of protecting the MCL along with ACLs during directional movement, and it affects the load of ACLs due to the decrease in MCL load. The MCL was reported to exhibit isometric movements upon knee flexion during the cutting maneuver, which is vulnerable to twisting or rotating movements due to a decrease in the degree of freedom in knee movement.[31,32] However, the knee movement performs internal rotation of the femur while the tibia is fixed through strengthening the hamstring, and the MCL reduces the contact cross-sectional area and pressure with the femur.[31,33] Therefore, hamstring muscles are important in reducing the load of ACL and MCL, and the load of ACL and MCL increases or decreases in proportion to each other during cutting maneuver.

The knee adduction and internal rotation moments during cutting maneuver were lower in the hamstring model than the other models. In a previous study, where musculoskeletal modeling and simulation experiments were conducted, the knee adduction and internal rotation moments simultaneously increased prior to the musculoskeletal modeling.[34] And the moments of the front and horizontal planes were high when changing directions, the same as the results of this study.[8,35,36] Considering the results of this study and previous studies, if the

Table 3

Results of knee joint moment as strengthening difference during cutting.

							Unit: Nm/(BW*HT)
	Abductor/hamstring	**Abductor**	**Hamstring**	**Control**	***F***	***P***	**Cohen's *d***
Flexion	−0.405 ± 0.611	−0.484 ± 0.333	−0.543 ± 0.297	−0.493 ± 0.495	1.002	.380	0.25
Extension	1.693 ± 0.701	1.636 ± 0.808	1.520 ± 0.709	1.638 ± 0.789	1.616	.202	0.40
Adduction	0.694 ± 0.438	0.773 ± 0.502†	0.616 ± 0.404‡	0.852 ± 0.716	3.775	**.028***	0.88
Abduction	−0.230 ± 0.562	−0.176 ± 0.308	−0.289 ± 0.269	−0.142 ± 0.334	2.285	.125	0.55
Internal rotation	0.247 ± 0.226	0.281 ± 0.247†	0.208 ± 0.190‡	0.257 ± 0.240	3.698	**.014***	0.87
External rotation	−0.254 ± 0.308	−0.227 ± 0.238	−0.298 ± 0.270	−0.233 ± 0.253	1.708	.190	0.42

*$P < .05$.

†Significant difference between abductor and hamstring.

‡Significant difference between hamstring and control.

moment of adduction and internal rotation of the knee joint decreases, it affects ACL and MCL load reduction. In addition, because the tension increases due to the strengthening of the hamstring muscles, the hip joint brings the knee to a neutral alignment when changing direction, and the load on the knee can be distributed to the hip joint.[8,34,37] However, the high moments of the frontal and horizontal planes cause high activation of the tensor fasciae latae and sartorius muscle in the abductor muscle, which increases the pressure on the ACL and meniscus, resulting in knee joint instability.[38] Through this study, hamstring strengthening was shown to directly reduce the knee joint moment during cutting maneuver. It is thus necessary to enhance muscle strength to a level optimized for performance through systematic training.

The load on the knee ligament was shown to be reduced by hamstring strengthening. The effects of increasing the hip abductor contraction force or the hip abductor and hamstring contraction forces together were not shown to be significant upon cutting maneuver. In reference to previous studies reporting a reduced risk of ACL injury in the cutting maneuver when the hip abductor muscle strength is increased, the hip abductor modeling was conducted.[39] However, the hip abductor muscle group and the surrounding muscles; rectus femoris, vastus lateralis and vastus medialis, simultaneously increased in strength through the training program to enhance the knee ligament stability in previous studies.[40] The lack of the strengthening of the hip abductor contraction force in this study may be due to the concurrent strengthening of the hip abductor muscles and the quadriceps muscles with rectus femoris, vastus lateralis, and vastus medialis through the training program.[10,41] It is also presumed that the knee stability fell due to the relative reduction of hamstring functions by the hip abductor contraction force.[29] Therefore, the hamstring contraction in the abductor/hamstring model increased, but the quadriceps contraction also increased. There was no effect of reducing the load on the knee joint.

The present study has several limitations. First, a limitation of this study is that an increase in the lower limb muscle training by 30% would be possible through simulation only. Nevertheless, the focus of such simulation-based analysis is on the sharing of information related to a decrease of knee ligament injuries through hamstring strengthening. Second, the force of the modeling muscle has limitations in generalizing to everyone because it applies the cadaver data used in previous studies. Third, based on the results of the muscle strengthening simulation in this study, further studies should conduct musculoskeletal modeling with a level of strengthening other than 30% as part of the continuous research with beneficial outcomes towards the highest performance of many athletes and a decrease of knee joint injuries. Fourth, cutting speed was the only performance demand for the side-cutting task. Other task demands, such as the anticipated vs. unanticipated nature of the task and muscle fatigue, were not evaluated and could interact with cutting speed to alter the movement patterns in the lower extremity. Therefore, it is necessary to consider these factors in future studies, minimize various variables, and measure them under the same conditions. Fifth, we assessed only lower extremity biomechanics previously shown to be associated with knee load. By analyzing kinematic factors such as knee flexion angle and angular velocity, the load according to the movement of the knee joint can be evaluated.

5. Conclusion

The aACL, iMCL, and pMCL forces during cutting maneuver were lower in the *hamstring* model than in the other models. In addition, the adduction and internal rotation moments of knee joint were lower in the *hamstring* model than in the other models. Incorporating the strengthening of the hip abductor muscle resulted in an augmented ratio of abductor contraction force relative to hamstring muscle strength. However, this enhancement was not associated with a reduction in knee joint load during cutting maneuvers. Interestingly, it was primarily the enhancement of the hamstring muscle that indicated a potential reduction in the load on the ACL and MCL. After undergoing ACL reconstruction surgery resulting from a rupture of ligament, athletes frequently encounter difficulties in returning to their prior level of sports activity.[42] According to the results of this study, coaches and trainers educate athletes who frequently perform cutting maneuver should be educated on the importance of hamstring muscles, and athletes are provided with injury prevention training programs related to hamstring muscles to improve performance by preventing knee ligaments.

Author contributions

Conceptualization: Sungmin Kim, Jeheon Moon.
Data curation: Jeheon Moon.
Funding acquisition: Jeheon Moon.
Investigation: Sungmin Kim, Seongyong Kim.
Resources: Sanghyuk Han, Seongyong Kim.
Visualization: Sungmin Kim.
Writing – original draft: Sungmin Kim, Jeheon Moon.
Writing – review & editing: Sanghyuk Han, Jeheon Moon.

References

[1] Kernozek TW, Ragan RJ. Estimation of anterior cruciate ligament tension from inverse dynamics data and electromyography in females during drop landing. Clin Biomech (Bristol, Avon). 2008;23:1279–86.
[2] O'Connor KM, Monteiro SK, Hoelker IA. Comparison of selected lateral cutting activities used to assess ACL injury risk. J Appl Biomech. 2009;25:9–21.
[3] Boden BP, Sheehan FT, Torg JS, et al. Non-contact ACL injuries_mechanisms and risk factors. Am Acad Orthop Surg. 2010;18:520–7.
[4] David S, Komnik I, Peters M, et al. Identification and risk estimation of movement strategies during cutting maneuvers. J Sci Med Sport. 2017;20:1075–80.

[5] Nasseri A, Khataee H, Bryant AL, et al. Modelling the loading mechanics of anterior cruciate ligament. Comput Methods Programs Biomed. 2020;184:105098.
[6] Andrews K, Lu A, McKean L, et al. Medial collateral ligament injuries. J Orthop. 2017;14:550–4.
[7] Kristianslund E, Faul O, Bahr R, et al. Sidestep cutting technique and knee abduction loading_implications for ACL prevention exercises. Br J Sports Med. 2014;48:779–83.
[8] Lee J, Pathak P, Panday SB, et al. Effect of foot-planting strategy on anterior cruciate ligament loading in women during a direction diversion maneuver: a musculoskeletal modeling approach. Orthop J Sports Med. 2020;8:2325967120963180.
[9] Moon J, Lee J, Kim K, et al. Effect of muscle-specific fatigue on the risk of anterior cruciate ligament injury in females. Appl Sci. 2021;11:4969.
[10] Zebis MK, Andersen LL, Ellingsgaard H, et al. Rapid hamstring quadriceps force capacity in male vs female elite soccer players. J Strength Cond. 2011;25:1989–93.
[11] Zillmer DA, Powell JW, Albright JP. Gender-specific injury patterns in high school varsity basketball. J Women's Health. 1992;1:69–76.
[12] Mueller MJ, Host JV, Norton BJ. Navicular drop as a composite measure of excessive pronation. J Am Podiatr Med Assoc. 1993;83:198–202.
[13] Weinhandl JT, Earl-Boehm JE, Ebersole KT, et al. Anticipatory effects on anterior cruciate ligament loading during sidestep cutting. Clin Biomech (Bristol, Avon). 2013;28:655–63.
[14] Cappozzo A, Cappello A, Croce UD, et al. Surface-marker cluster design criteria for 3-D bone movement reconstruction. IEEE Trans Biomed Eng. 1997;44:1165–74.
[15] Wu G, Siegler S, Allard P, et al.; Standardization and Terminology Committee of the International Society of Biomechanics. ISB recommendation on definitions of joint coordinate system of various joints for the reporting of human joint motion-part I: ankle, hip, and spine. J Biomech. 2002;35:543–8.
[16] Donnelly CJ, Lloyd DG, Elliott BC, et al. Optimizing whole-body kinematics to minimize valgus knee loading during sidestepping: implications for ACL injury risk. J Biomech. 2012;45:1491–7.
[17] McErlain-Naylor S, King M, Pain MTG. Determinants of countermovement jump performance_a kinetic and kinematic analysis. J Sports Sci. 2014;32:1805–12.
[18] Beaulieu ML, Lamontagne M, Xu L. Lower limb muscle activity and kinematics of an unanticipated cutting manoeuvre: a gender comparison. Knee Surg Sports Traumatol Arthrosc. 2009;17:968–76.
[19] Sikidar A, Marieswaran M, Kalyanasundaram D. Estimation of forces on anterior cruciate ligament in dynamic activities. Biomech Model Mechanobiol. 2021;20:1533–46.
[20] Delp SL, Loan JP, Hoy MG, et al. An Interactive graphics based model of the lower extremity to study orthopaedic surgical procedures. IEEE Trans Biomed Eng. 1990;37:757–67.
[21] Sutter R, Kalberer F, Binkert CA, et al. Abductor tendon tears are associated with hypertrophy of the tensor fasciae latae muscle. Skeletal Radiol. 2013;42:627–33.
[22] van den Tillaar R, Solheim JAB, Bencke J. Comparison of hamstring muscle activation during high-speed running and various hamstring strengthening exercises. Int J Sports Phys Ther. 2017;12:718–27.
[23] Xu H, Bloswick D, Merryweather A. An improved OpenSim gait model with multiple degrees of freedom knee joint and knee ligaments. Comput Methods Biomech Biomed Engin. 2015;18:1217–24.
[24] Myers CA, Laz PJ, Shelburne KB, et al. Simulated hip abductor strengthening reduces peak joint contact forces in patients with total hip arthroplasty. J Biomech. 2019;93:18–27.
[25] Delp SL, Anderson FC, Arnold AS, et al. OpenSim_open-source software to create and analyze dynamic simulations of movement. IEEE Trans Biomed Eng. 2007;54:1940–50.
[26] Kar J, Quesada PM. A musculoskeletal modeling approach for estimating anterior cruciate ligament strains and knee anterior-posterior shear forces in stop-jumps performed by young recreational female athletes. Ann Biomed Eng. 2013;41:338–48.
[27] Lu Y, Mei Q, Peng H-T, et al. A comparative study on loadings of the lower extremity during deep squat in Asian and Caucasian individuals via OpenSim musculoskeletal modelling. Biomed Res Int. 2020;2020:1–10.
[28] Simonsen EB, Magnusson SP, Bencke J, et al. Can the hamstring muscles protect the anterior cruciate ligament during a side-cutting maneuver. Scand J Med Sci Sports. 2000;10:78–84.
[29] Kong PW, Burns SF. Bilateral difference in hamstrings to quadriceps ratio in healthy males and females. Phys Ther Sport. 2010;11:12–7.
[30] Shin CS, Chaudhari AM, Andriacchi TP. The influence of deceleration forces on ACL strain during single-leg landing: a simulation study. J Biomech. 2007;40:1145–52.
[31] Hosseini Nasab SH, Smith CR, Schutz P, et al. Elongation patterns of the collateral ligaments after total knee arthroplasty are dominated by the knee flexion angle. Front Bioeng Biotechnol. 2019;7:323.
[32] Feeley BT, Muller MS, Allen AA, et al. Biomechanical comparison of medial collateral ligament reconstructions using computer-assisted navigation. Am J Sports Med. 2009;37:1123–30.
[33] Dennis DA, Komistek RD, Mahfouz MR, et al. Multicenter determination of in vivo kinematics after total knee arthroplasty. Orthop Relat Res. 2003;416:37–57.
[34] Gupta D, Reinbolt JA, Donnelly CJ. Optimizing whole-body kinematics during single-leg jump landing to reduce peak abduction/adduction and internal rotation knee moments: implications for anterior cruciate ligament injury risk. J Appl Biomech. 2021;37:432–9.
[35] Nedergaard NJ, Dalbo S, Petersen SV, et al. Biomechanical and neuromuscular comparison of single- and multi-planar jump tests and a side-cutting maneuver: Implications for ACL injury risk assessment. Knee. 2020;27:324–33.
[36] Witvrouw E, Cambier D, Danneels L, et al. The effect of exercise regimens on reflex response time of the vasti muscles in patients with anterior knee pain: a prospective randomized intervention study. Scand J Med Sci Sports. 2003;13:251–8.
[37] Park SE, DeFrate LE, Suggs JF, et al. Erratum to "The change in length of the medial and lateral collateral ligaments during in vivo knee flexion". Knee. 2006;13:77–82.
[38] Besier TF, Lloyd DG, Ackland TR. Muscle activation strategies at the knee during running and cutting maneuvers. Med Sci Sports Exerc. 2003;35:119–27.
[39] McLean SG, Huang X, Su A, et al. Combined knee loading states that generate high anterior cruciate ligament forces. Clin Biomech. 2004;19:828–38.
[40] McCurdy K, O'Kelley E, Kutz M, et al. Comparison of lower extremity EMG Between the 2-leg squat and modified single-leg squat in female athletes. J Sport Rehabil. 2010;19:57–70.
[41] Petersen W, Ellermann A, Gosele-Koppenburg A, et al. Patellofemoral pain syndrome. Knee Surg Sports Traumatol Arthrosc. 2014;22:2264–74.
[42] D'Ambrosi R, Meena A, Raj A, et al. Multiple revision anterior cruciate ligament reconstruction: not the best but still good. Knee Surg Sports Traumatol Arthrosc. 2023;31:559–71.